

AUM VEKARIYA

Long Beach, CA - 90815 | +1 (562) 277-5369 | AI/ML Engineer, MS Student | Open to Relocation
aumghanshyam.vekariya01@student.csulb.edu | linkedin.com/in/aum-vekariya | aumvekariya.vercel.app

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Hugging Face Transformers, LangChain, LangGraph, CrewAI, LlamaIndex, RAG Pipelines, Prompt Engineering, Vector Databases (Qdrant, pgvector), Langfuse, MLflow

Backend & Real-Time: FastAPI, REST APIs, WebSockets, WebRTC (LiveKit), gRPC, Kafka

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, Lambda, GPU), Jenkins, CI/CD, Linux

Frontend: React.js, Next.js, Redux, Tailwind CSS, TypeScript

Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis, Stripe

EXPERIENCE

Software Developer (AI/ML) | Zluck Solutions — Surat, Gujarat, India

Mar 2025 – Sep 2025

- Reduced average API response latency by **25%** across AI-assisted endpoints by refactoring **MySQL/MongoDB** query paths, adding composite indexes, and fixing repeated database calls on high-traffic service routes.
- Improved backend decision-flow consistency by **30%** and reduced processing costs by **~\$10K/quarter** by integrating **LLM-based classification and summarization agents (LangChain)**, replacing brittle rule chains with prompt-driven inference pipelines.
- Eliminated **3+ recurring regression classes** by automating CI/CD pipelines with **AWS S3/Lambda**, integrating structured logging and test coverage gates, enabling weekly deployments without production rollbacks.

Machine Learning Intern | KrtrimaIQ Cognitive Solutions — Bengaluru, Karnataka, India

Dec 2024 – Mar 2025

- Reduced manual teacher workload by **45%** by designing and deploying **LLM-driven content generation and essay-scoring services** using **FastAPI, LangChain, and OpenAI APIs**, automating assignment creation and rubric-based evaluation at scale.
- Improved LLM output contextual relevance by **35%** by implementing real-time **RAG evaluation with Qdrant** vector retrieval and WebSocket streaming, grounding responses on curriculum-aligned passages and halving off-topic generation.
- Eliminated manual classroom document prep by automating **multilingual PDF generation** (Hindi/English) with **PyLaTeX/XeLaTeX** and prompt-tuned agents, producing print-ready output at scale with near-zero formatting defects.

Software Developer Intern (Co-op) | Zluck Solutions — Surat, Gujarat, India

Dec 2023 – May 2024

- Reduced client-reported API issues by **15%** by implementing request validation, standardized error contracts, and request/response test suites across **4+ production applications** built with **Python, SQL, and JavaScript**.
- Strengthened business-logic correctness on high-traffic paths by applying DSA-driven data transformations and systematic boundary-condition handling, measurably reducing failure rates in production-grade code.
- Built foundational ML fluency by training NLP and CV models using **Hugging Face Transformers and PyTorch**, prototyping a text classification module later adopted as a proof-of-concept for the team's AI roadmap.

PROJECTS

Real-Time AI Voice Assistant for Sales Training | Python · WebRTC (LiveKit) · Whisper · GPT-4o · KokoroTTS · Docker

- Achieved **sub-500ms end-to-end turn latency** by engineering a full **STT → LLM → TTS** streaming pipeline over **WebRTC (LiveKit)** with barge-in interruption handling, enabling real-time persona-driven sales simulations across **5 distinct archetypes**.

AI-Powered EdTech Platform for Academic Workflow Automation | FastAPI · LangChain · Qdrant · WebSockets · PyLaTeX · AWS

- Cut manual curriculum document creation time by **60%** by building a multilingual **RAG pipeline (LangChain + Qdrant)** with real-time approval workflows over **WebSockets**, delivering print-ready PDFs via **PyLaTeX** on AWS.

Vaultspace: Multi-Tenant AI Workspace | FastAPI · PostgreSQL/pgvector · React · TypeScript · Tailwind · Stripe · OpenAI

- Built a full-stack multi-tenant SaaS workspace with **Postgres RLS** tenant isolation, **RBAC**, **JWT** auth, **Stripe** billing, a per-tenant **pgvector** RAG assistant, and a 24-test pytest suite proving cross-tenant data cannot leak.

EDUCATION

California State University Long Beach

Sep 2025 – May 2027

Master of Science in Computer Science

Long Beach, CA

Coursework: Advanced Software Engineering, Advanced Algorithms, Distributed Systems, Advanced Artificial Intelligence, Pattern Recognition, Computer Vision, Programming Languages, Search Engine Technology.

Charotar University of Science and Technology

Jun 2021 – May 2025

Bachelor of Technology in Computer Engineering

Changa, India

Coursework: Software Engineering, Data Structures and Algorithms, Operating Systems, Database Management System, Blockchain Technology, Computer Networks, Internet of Things, Machine Learning, Artificial Intelligence, Network Security.

PUBLICATIONS

Authored and published a peer-reviewed paper on decentralized financial audit systems, *Decentralized Distributed Financial Audit System*, accepted at ADCIS 2024 (Springer, LNNS vol. 1333).